THE WITNESS Environment Kit Development Roadmap

Updated 9 Nov 2026 · Revision 3 · AI-led production · Q01 required, L04 extension · Quality gates replace the fixed eight-week deadline. Three.js r187. Phases 1–2 now have regional references, compressed PBR assets, lifecycle/offline evidence and a laptop GPU comparison. The original document is retained as a backup.

1. Goal, scope and assumptions

Goal. Build an authored, reusable Environment Kit and architecture that can actually be explored. Q01 must support a continuous playable route through a doorway, an interior, stairs and an upper floor or roof, and back to the harbor. Visual quality is judged at player eye height and during movement as well as against the cover-art reference.

Required slice and extension:

- **Q01 "Bến không chờ người về"** (Caesarea harbor) — the first scene every player sees.
- L04 Galilee shore (Ga 21) is an extension after Q01 meets its gates. It adds basalt, water, boats and nets. It is not a second mandatory scene for an eight-week deadline.

Jerusalem, L01 Exodus, L02 En Gedi, L03 cistern and L05 prison remain backlog. Reuse is an architectural goal, not a claim that those scenes are already kit-ready or dressed.

Confirmed production decisions:

- AI performs all build work: scripts, asset generation, conversion, integration, tests and repairs. No human modelling or programming labor is assumed. The old 15–20 hours/week and 120–160 human-hour estimate is withdrawn.
- Quality takes priority over a fixed completion date. Each gate requires evidence. AI generation speed does not by itself certify visual quality, historical accuracy or device performance.
- The game remains on Three.js. Editable sources and glTF/PBR support reuse; a future engine port would still require gameplay and integration work.
- Existing gameplay must be preserved. Run model/source, browser, craft and graphics checks for relevant runtime changes. The graphics and model baseline passed; the full browser campaign suite was not completed in the preceding graphics change.

Outside the current scope: new character art, mocap, WebGPU migration, Temple Mount, full Legends dressing and cinematic VFX rework. The existing L01 fire and sea shaders are retained.

2. Starting point: what the repo tells us

These findings drive the plan.

Finding	Where	Consequence for the kit
Shared-core GLTFLoader, KTX2Loader and MeshoptDecoder are bundled against Three r187. GTAO already exists.	vendor/witness-loaders.js and engine core	Reuse the tested pipeline and GTAO. Scene-owned scopes handle abort and shared-resource release.
Bundled Three.js is r187 and npm `three@0.187.0` is the latest release and contains every needed addon	engine bundle; npm registry	Pin addons to exactly 0.187.0 so they match the bundle
`BatchedMesh` and `InstancedMesh` already exist in the core	engine bundle	Use them for repeated modules and vegetation
Buildings are ~20 stacked boxes created by `house()`; boats use lathe + boxes; the lighthouse is stacked cylinders	`witness-world.js`	These are the first things the kit replaces
`height(x,z)` ray-casts the ground mesh for every placed object	`witness-world.js` `decorate()`	Profile placement cost; share a final-geometry sampler with terrain support.
Collision uses circle `solids` registered per building; interaction targets must not get solids	`house()`, `applySolids()`	Kit pieces need collision footprints in metadata and must keep NPC/prop coordinates stable
Terrain already uses four PBR layers with AO/roughness/height atlases. Architecture remains mainly procedural.	`vendor/adventure-visuals.js`	Extend material families and authored terrain; do not repeat the initial terrain conversion.
Player standalone embeds the active lighter pack. A separate lab standalone proves full KTX2/Meshopt and bundled WASM without HTTP.	`tools/build.cjs`	Keep immutable asset URLs, refreshable manifests and successful cache writes. Actual service-worker offline reload is tested.
Single ground plane (156 m wide), playable area about 78 × 66 m per scene	`witness-world.js`	Budgets are per scene, which keeps them tractable
noSurface/noCast were removed before material construction.	`mat()` in `witness-world.js`	Keep regression coverage; this warning fix is complete.

3. Look Bible and evidence rules

Use the existing project evidence vocabulary: [K] Scripture, [L] historical/archaeological source, [G] Catholic teaching, [H] artistic invention, [C] uncertain claim. The original roadmap labels [V] and [D] map to [L] and [H]. Record sources, dates, confidence and reconstruction boundaries per claim.

Target look: assets/journey-cover.webp guides warm light, weathered surfaces, muted earth colours and atmospheric depth. The regional board now records eight source observations per region with original diagrams and external source-image links; six reproducible pre-kit game views per scene provide the comparison baseline. Archaeological photographs are not redistributed without verified rights.

Material families per region are art-direction choices. Mark historically supported claims [L] and distinguish reconstruction [H/C].

Region / scene	Primary materials	Key facts to respect
Caesarea harbor (Q01)	Pale local **kurkar** sandstone and limestone ashlar, imported marble accents, lime plaster, hydraulic concrete (opus caementicium) on the moles, timber, rope	Kurkar/limestone, plaster, timber and harbor construction are references [L]. Distinguish construction phases from the documented inauguration in 10 BCE. Exact 8 cm / 2 cm dressing dimensions and lighthouse reconstruction require asset-specific sources [C/H].
Galilee (L04, Capernaum-type village)	Local basalt, coarse and irregular; mud/earth joints; timber and leaves with pressed-mud roof surfaces. Plaster and later repairs require phase-specific evidence.	Custodia describes basalt houses with mud/earth mortar and timber/leaves/pressed-mud roofs. Do not impose a universal dry-stone rule. Capernaum is comparative, not the established location of Ga 21. Museum boat data is comparative; final dimensions need excavation plans.
Jerusalem (later)	Pale limestone, Herodian ashlar, narrow stepped streets	Reuse Caesarea masonry with different weathering
L01 Exodus	Sand, dry earth, goat-hair tents, reeds, cart wreckage	Mostly nature and soft materials [H]
L02 En Gedi	Cliff limestone, caves, springs, thorny scrub	Rock kit and cave pieces [H]
L03 cistern, L05 prison	Cut limestone, wet stone, timber, iron	Interior pieces; keep existing gameplay layout

Palette rule. No neon or modern saturation; earth tones only, as in your character research (madder red, woad blue, ochre, undyed linen and wool). Sacred Tier presence is shown through light, not extra props.

Region locks control art direction and prevent accidental reuse. They are not absolute archaeological claims that a material could never occur outside its preferred region. Use terrain, season and habitat when selecting vegetation.

4. Technical standards in docs/environment-kit/standards.md

Item	Standard
Units and axes	1 Blender unit = 1 m; Y up on export; pivot at bottom centre of each module
Module grid	Architecture on a 0.5 m grid; wall modules 2 m and 4 m wide, 3 m and 4.5 m tall
Texel density	Terrain 256 px/m; architecture and props 512 px/m; hero props up to 1024 px/m
Texture sets	Albedo (sRGB), normal, ORM packed (occlusion / roughness / metal). Source 2K, ship 1K–2K as KTX2 (UASTC for normals, ETC1S for albedo/ORM)
Triangle budgets	Wall module ≤ 800; column ≤ 1,500; small prop ≤ 3,000; hero prop ≤ 12,000; tree LOD0 ≤ 8,000; rock ≤ 2,500
LODs	LOD where repetition, screen size and measured scene cost justify it; prioritize large repeated assets and vegetation.
Materials	One material per module (trim-sheet or atlas) so draw calls stay low
Vertex colour	AO/wear/wind masks only with an explicit shader contract; ordinary RGB vertex colour remains separate.
Collision	Local extras.footprint circles transformed per node/instance for 2D obstacles; separate 3D collision and support surfaces for accessible buildings.
Naming	`ENV_<region>_<type>_<variant>_LOD0`, e.g. `ENV_CAE_Wall_Ashlar_4m_A_LOD0`
Licence	Every external asset gets a row in `assets/ART-CREDITS.md`: source URL, licence, date, modifications

Per-scene budgets remain provisional targets: desktop 1.2 M visible triangles, 250 draw calls and 256 MiB texture memory; mobile balanced 400 k, 120 and 128 MiB. A dedicated 200-module scene is measured on this laptop's Intel Arc; headless GPU time and texture upload estimates do not certify full-scene FPS or driver residency.

5. Toolchain and asset sources

Purpose	Tool / source	Notes
Modelling, UV, bake	Blender (free)	Use Geometry Nodes for stone-block and thatch generators; bake AO to vertex colour
Optimise glTF	`@gltf-transform/cli` (4.5.1 on npm)	dedup, weld, simplify, meshopt, texture resize
Texture compression	Basis Universal v2.50 WASI	Pinned encoder with SHA256 verification; ETC1S albedo/ORM and UASTC normal, ten mip levels.
Loader bundle	`esbuild` + `three@0.187.0` addons	Shared-core GLTF/KTX2 plus Meshopt, owned scene scopes and bundled codecs are tested. Existing GTAO/grade/fog/shadow work is retained.
CC0 textures, models, HDRIs	**Poly Haven**, **ambientCG**	Poly Haven states all assets are CC0 and usable commercially; ambientCG is CC0 too. Both are safe to ship, but still record each in the ledger
Scanned assets	**Fab / Megascans**	The Fab Standard License allows use in any engine, but pricing is per asset and the EULA has distribution limits. Read it before shipping extractable files in a browser game
One-off models	Sketchfab	Per-model licence; accept only CC0 or CC-BY with credit; never "editorial"
Blockout only	Kenney, Quaternius	Stylised CC0; do not ship in the final look

6. Milestone plan and acceptance gates

Phase 1 Foundations and baseline

Create the Look Bible, standards, catalogue and repeatable Q01/L04 views. Define required Q01 access and distinguish measured device results from emulated checks.

Phase 2 AI asset pipeline

Prove source-to-GLB generation, shared-core loading, instancing, transformed footprints, resource ownership and offline packaging. KTX2, Meshopt and locally bundled codecs now have actual browser tests; retain these checks as the pipeline expands.

Phase 3 Playable architecture blockout

Build harbor to doorway to room to stairs to upper floor or roof and back. Implement 3D support/collision and floor-aware navigation and interaction. Save and resume on the correct floor. Verify first-person and third-person cameras.

Phase 4 Q01 look development

Finish one playable corner with ground, wall base, doorway, quay and props. Review scale, texture repetition, module seams, wear and light at eye height. Do not accept only a beauty screenshot.

Phase 5 Complete Q01

Create only the modules and nature/prop assets needed for the required route and scene framing. Preserve quest reachability and validate each replacement before removing its fallback.

Phase 6 Optimisation and lighter offline build

Tune LOD, textures, foliage and effects. Keep the same essential walkable interiors, stairs and roof access in the lighter build. Test fresh load, save/resume, scene transitions and missing assets with network blocked.

Phase 7 L04 extension

After Q01 passes, extend the kit with basalt, shore, boat and net assets. Define boat boarding or moving-platform behavior separately if included. Preserve Q01 quality.

Phase 8 Final evidence and delivery

Run functional and visual checks, static budgets and device-specific profiling. Store evidence against source/build revisions. Record remaining limits. Other scenes remain backlog.

7. Asset catalog (priority checklist)

Priority	Group	Items (approx. count)
P0	Materials	13 candidate families; produce only required Q01 families first
P0	Terrain	Q01 authored heightmap and path masks; L04 terrain as extension
P0	Caesarea architecture	About 25 candidate modules; build only those needed for Q01 first
EXT	Galilee architecture	~15 modules incl. courtyard, stairs, thatch roof, basalt quay
P0	Nature	Only nature assets required by Q01 framing first
P0	Hero props	Q01 hero needs first; boat for L04 extension; tent and windlass as needed by later scenes
P1	Small props	~25 pieces
P1	Decals	~20 (dirt, stains, cracks, moss, marks)

Priority	Group	Items (approx. count)
P2	Jerusalem street set	~15 modules
P2	Cistern and prison interiors	~15 modules (replaces the open cylinder in L03 and the box prison in L05)
P2	En Gedi cave pieces	~6

Every catalog entry records its source, region, evidence, licence, priority, collision/access contract, reproducible source, art acceptance and validation evidence. A pipeline sample does not count as final art.

8. Integration architecture

- Loader bundle. WitnessKit.create(T, renderer) creates an owned runtime; init remains available for legacy callers. Addons share engine Three r187. The PBR study has PNG reference, KTX2 plus Meshopt, and a lighter PNG tier. Exact JS/WASM decoders ship locally. Source scripts and encoder checksums make conversion repeatable.
- Manifest-driven scenes. Load with scene-generation/cancellation guards, stable placeholders and explicit ownership of shared resources. Keep existing builders until replacement visuals, collision and gameplay are verified. Do not append an asset to a scene that has already been disposed.
- Height and support. Use a shared terrain sampler derived from the final terrain geometry, including seabed changes. Multi-level buildings additionally need 3D support surfaces and collision: a single height(x,z) cannot represent both a room floor and its roof.
- Collision. Node-local footprints use node and instance transforms. Compressed assets declare footprintSpace=asset so quantized geometry transforms cannot distort authored collision. Footprints remain 2D obstacles. Add 3D walls, floors, ceilings, stairs, camera collision and floor-aware interactions in Phase 3.
- Materials. Reuse current PBR terrain. Build only required material families first. Keep shader overlays for documented dust/wear effects. Do not reinterpret the existing AO/roughness/height atlas as a standard metallic ORM map.
- Packs and offline. Use versioned region packs and runtime caching with actual cache writes and failure handling. The lighter standalone build may reduce textures, detail and effects, but retains the same essential accessible architecture and progress semantics. Test without network, including decoder/transcoder dependencies.
- Shadows and atmosphere. Follow-player 44 x 44 m shadows, GTAO, scene LUTs, height fog and up-to-4x render-target MSAA are already integrated. Tune these against new assets; do not schedule a second implementation without evidence.

9. QA gates and metrics

- Functional acceptance: relevant npm test, npm run test:browser, npm run test:craft-details and npm run test:graphics checks. Add route tests for doors, stairs, floors, ceilings, roofs, camera clipping, save/resume and interactions across levels. Missing evidence keeps a gate open.

- Visual baseline: six recorded views each of Q01 and L04 before kit dressing, plus first-person and traversal checks. Store camera/player parameters and fixture identity with images so comparisons can be repeated.
- Performance: specify actual device, resolution, quality and test duration. Desktop 60 fps and mobile balanced 30 fps remain targets. Record frame-time distribution and repeated scene changes; headless SwiftShader is functional QA, not a phone performance measurement.
- Budget checks: test/kit-budget.cjs checks pack/catalog consistency and static GLB budgets during build. Runtime draw calls, residency, render targets and FPS require separate instrumentation; static asset checks cannot certify them.
- **Licence audit:** every external asset has a ledger row before it is committed.

10. Risks and mitigations

Risk	Mitigation
Scope creep	Q01 required; L04 extension; expand only after playable and visual gates pass.
Modelling takes longer than planned	Reproducible AI scripts, early reference corner and smaller catalogue. No human modelling labor is assumed.
Loader and Three.js version mismatch	Pin r187, share the engine core, retain actual GLB/KTX2/Meshopt browser tests.
Offline HTML build bloats	Versioned packs and a lighter build preserving essential playable architecture.
Performance on phones	Balanced tier with smaller textures, fewer instances, no decals
Historical or tonal inconsistency	Look Bible, region locks, [K]/[L]/[H] tags
License trouble	Prefer Poly Haven and ambientCG; ledger every asset; read Fab EULA before using scans
Breaking gameplay coordinates	Preserve anchors and verify reachable paths, floor-aware interaction and quest regression after each replacement.

11. Scope decisions and gate evidence

- Gate 0: standards, source ledger and repeatable visual baseline. Gate 1: asset pipeline including loading, transformed collision metadata, texture compression and offline dependencies.
- Gate 2: a blockout route through an interior, stairs and a roof works in both camera modes, with correct collision, save/resume and floor-aware interactions.
- Gate 3: one playable Q01 look-development corner meets the visual standard. Expand Q01 only after this corner is accepted.

- Gate 4: complete and optimize Q01, validate the lighter offline build, then decide whether to extend L04. A gate stays open when any required test or quality evidence is missing.

If scope must be reduced, reduce catalog breadth and defer L04 first. Keep the playable Q01 architecture and its quality. Do not make a late week-five decision to claim a one-month schedule.

12. Immediate development work

1. Use docs/environment-kit/roadmap.md as the working roadmap, standards.md as the asset contract, look-bible.md for visual/evidence direction, catalog.csv for asset records and progress.json for gate evidence. Update these with each accepted build.
2. Phases 1–2 evidence: twelve repeatable baseline views; sixteen annotated regional observations from eight sources; a 288-triangle PBR module in three quality tiers; hash/mip/colour-space checks; abort, shared-resource and repeated scene-change tests; actual runtime-cache offline reload and a cold standalone decoder test.
3. Next: Phase 3 playable blockout, harbor to doorway to room to stairs to roof and back. No final architecture, stairs, art corner or physical-phone performance is accepted by the pipeline test. No human modelling or programming step is a prerequisite.

13 Sources and change record

Revision 3 records Phase 1–2 implementation evidence. The 200-module Intel Arc comparison uses a 1920 × 1080 drawing buffer, two repeats per quality tier and GPU timer queries. It excludes display presentation and human input; it is not a phone or complete-Q01 result. Read validation/hardware-profile.json for exact numbers and conditions.

Working documents: docs/environment-kit/roadmap.md; standards.md; look-bible.md; catalog.csv; progress.json. Code and machine-readable evidence determine implementation status, not this prose alone.

Three.js GLTFLoader: https://threejs.org/docs/pages/GLTFLoader.html

Three.js KTX2Loader: https://threejs.org/docs/pages/KTX2Loader.html

Instancing and spatial culling: https://github.com/KhronosGroup/glTF/blob/main/extensions/2.0/Vendor/EXT_mesh_gpu_instancing/README.md

Caesarea park source: https://static-en.parks.org.il/wp-content/uploads/2026/02/ENG_Caesarea-Park.pdf

Capernaum source: https://www.custodia.org/en/sanctuaries/capernaum/

Research register: docs/environment-kit/research/references.json and source-register.md. The HTML board uses original illustrative diagrams; restricted source photographs are linked, not redistributed.

Shor et al. 2026: https://doi.org/10.1617/s11527-026-03228-9. Distinguish construction functions and plaster from exposed stone. Area LL storage includes later periods: https://hadashot.iaa.org.il/Report_Detail_Eng.aspx?id=25450.

PBR study source: https://polyhaven.com/a/coral_ground_02 (CC0). This tests porous limestone rendering, not exact Caesarea kurkar. The material and wall remain unaccepted final art.